AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 6 · STUDENT LESSON

Forward passes by hand

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

In Week 5, what does the output layer receive?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Record scores and activations in dependency order. Try the recall before opening notes.

Your learning target

I can record scores and activations in **dependency** order.

I can locate the first incorrect value in a forward **trace**.

I can separate a final **output** from **evidence** of model accuracy.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Record scores and activations in **dependency** order

Locate the first incorrect value in a forward **trace**

Separate a final **output** from **evidence** of model accuracy

Retrieve what you already know

> In Week 5, what does the **output** layer receive?

My starting explanation

This week's end goal

By the end, I can explain forward passes by hand using a **trace**d example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Forward passes by hand case. Separate observations, given facts and assumptions before choosing an explanation.

Use a new one-**input** network with two hidden units. Its scores are z1 = 2x - 1 and z2 = -x + 2. Apply ReLU to obtain h1 and h2. The **output** is y = h1 - 2h2 + 1. Parameters stay fixed while the **input** changes.

The **input** x is an abstract number; these are arithmetic examples, not measurements of children.

A forward pass evaluates this network in **dependency** order. Compute both scores from x, then both activations, then the **output** contributions h1 and -2h2, and finally add bias 1.

A **trace** table makes an omitted bias or a sign error easier to locate than a final answer alone.

x	z1 and z2	h1 and h2	y
0	-1 and 2	0 and 2	-3
1	1 and 1	1 and 1	0
3	5 and -1	5 and 0	6

At x = 1 the **output** is exactly zero. If we later declare class 1 for y >= 0, equality belongs to class 1. Applying ReLU in the hidden layer does not make the final **output** nonnegative.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Forward passes by hand. Connect each labelled input, deciding step and result.

For x = 0, the first score is -1, which ReLU changes to 0. The second score is 2, which ReLU leaves at 2. Therefore y = 0 - 4 + 1 = -3. Negative **output** is allowed because this **output** layer uses identity.

```
def trace(x):
    z = [2 * x - 1, -x + 2]
    h = [max(0, value) for value in z]
    y = h[0] - 2 * h[1] + 1
    return z, h, y

for x in [0, 1, 3]:
    print(x, trace(x))
```

This optional Python uses only numeric **input**s. Compare every returned component against the table.

The same code being run twice is a repeatability check; comparing it with a separately calculated expectation is a stronger check of correctness.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Forward passes by hand. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: A submitted x = 0 **trace** uses h=-1 and y = 5.

Predicted result

Prediction: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Why the result follows

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Forward passes by hand to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Trace	A record of intermediate values in a computation

Word	Meaning
Preactivation	A weighted score before an activation function
Dependency	An earlier value needed to compute a later value
Regression check	A retained test rerun after a change to detect a new error

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Forward passes by hand, predict a step, then compare with its explanation.

Correct a trace from the earliest error

TRACE STAGE AT X IS 0	EXPECTED VALUE	SUBMITTED VALUE
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Trace x = 2, recording both scores, both activations and y. What happens to the second unit at its zero-score boundary?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Forward passes by hand

Audit a forward pass by writing every value at its correct stage.

The facts we will use

Supplied facts

Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/6

Forward passes by hand

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preactivation

At x=0, z=-1; at x=2, z=3.



2. Activation

At x=0, h=0; at x=2, h=3.



3. Output

Outputs are 4 and 1 respectively.

CHANGE ONE THING

A submitted x = 0 trace uses h=-1 and y = 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Preactivation	At x=0, z=-1; at x=2, z=3.
2. Activation	At x=0, h=0; at x=2, h=3.
3. Output	Outputs are 4 and 1 respectively.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

What this example does not establish

Keep the stated limit

A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

Keep the stated limit

Read the labelled picture: Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A submitted x = 0 **trace** uses h=-1 and y = 5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Forward passes by hand claim and explain a repair using the supplied evidence.

Check the claim against the facts

A test where both scores are positive cannot reveal a missing ReLU. Include x = 0, where the first score is negative, and x = 3, where the second score is negative.

Include x = 0.5 and x = 2 to exercise each hidden zero-score boundary. Neither a few examples nor an attractive diagram proves every implementation case is correct.

A missing **output** bias shifts every **output** by -1. A swapped hidden order may pass a case where h1 equals h2; x = 1 has this limitation.

Keep several checks with different intermediate values, and retain them when repairing code.

Locate the earliest unsupported step

Trace x = 0.5 and x = 2. Which hidden score is zero in each case?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Forward passes by hand with support, then explain the deciding step in your own words.

Trace x = -1 through the lesson network. Record z, h and y.

Trace x = 2, recording both scores, both activations and y. What happens to the second unit at its zero-score boundary?

Explain your prediction to a partner before comparing results.

Practise with less help: Locate a bias omission

Use workbook W2 for the connected example “Forward passes by hand”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Forward passes by hand. Proposed explanation: “ReLU means the **output** cannot be negative”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Forward passes by hand. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

```
def trace(x):
    z = [2 * x - 1, -x + 2]
    h = [max(0, value) for value in z]
    y = h[0] - 2 * h[1] + 1
    return z, h, y

for x in [0, 1, 3]:
    print(x, trace(x))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, **trace** the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Forward passes by hand. Explain why each step is justified.

Explain → apply → challenge

Trace x = -1 through the lesson network. Record z, h and y.

> **Trace** x = 0.5 and x = 2. Which hidden score is zero in each case?

Explain why testing only x = 1 in the lesson model may miss a hidden-order swap.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled forward passes by hand **evidence** sheet to Image-pattern Network Investigation. Show how you can record scores and activations in **dependency** order.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a **trace** is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A forward **trace** verifies what specified parameters calculate for one **input**. It does not show how those parameters were learned or whether the **output** is useful.

Classification accuracy also needs a declared class rule, true labels and suitable independent examples.

A program matches y at x = 1 but its hidden activations are wrong. Is the implementation verified? Explain what evidence to retain after fixing it.

Next week separates the network architecture from its numerical parameters and counts the values that **training** could change.

Explain why testing only x = 1 in the lesson model may miss a hidden-order swap.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Forward passes by hand case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Correct a trace from the earliest error

TRACE STAGE AT X IS 0	EXPECTED VALUE	SUBMITTED VALUE
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit a different network independently

Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error and correct all later values.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error and correct all later values.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Forward passes by hand, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Audit a forward pass by writing every value at its correct stage.

By the end, I can	How I will show it
Record scores and activations in dependency order	Explain the deciding step, show a trace or test, and state one limit.
Locate the first incorrect value in a forward trace	Explain the deciding step, show a trace or test, and state one limit.
Separate a final output from evidence of model accuracy	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Forward passes by hand without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Record scores and activations in **dependency** order

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Audit a forward pass by writing every value at its correct stage.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Output**s are 4 and 1 respectively.

1. **Preactivation**

2. **Output**

3. Activation

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Output Output**s are 4 and 1 respectively. A forward pass applies a fixed network to an **input**; it does not update weights.

Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A submitted x = 0 **trace** uses h=-1 and y = 5. Which conclusion is supported by the supplied facts?

1. Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4. ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Repair a missing building block

Review Composing layers (ai-5/5). Goal: **Trace** how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a **trace** rather than an execution.

Keep your completed work

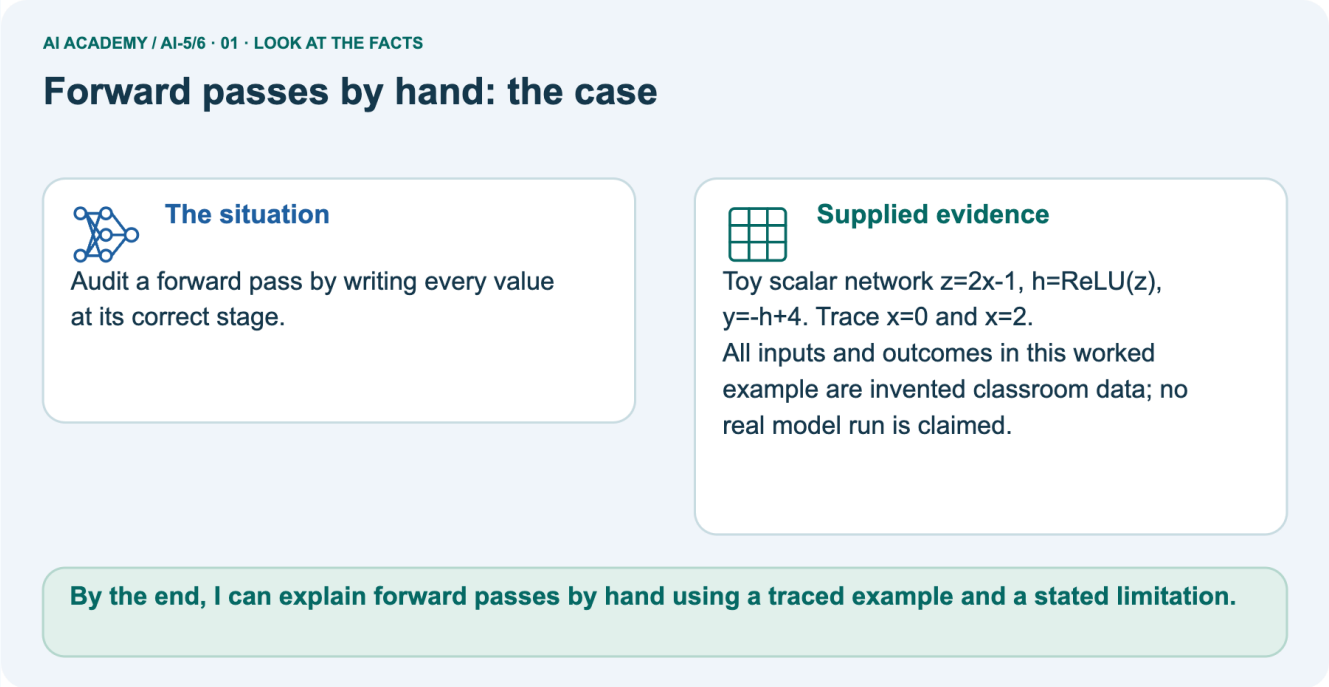
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Forward passes by hand graphic using its labels, deciding evidence and relevant limitation.

01. Forward passes by hand: the case



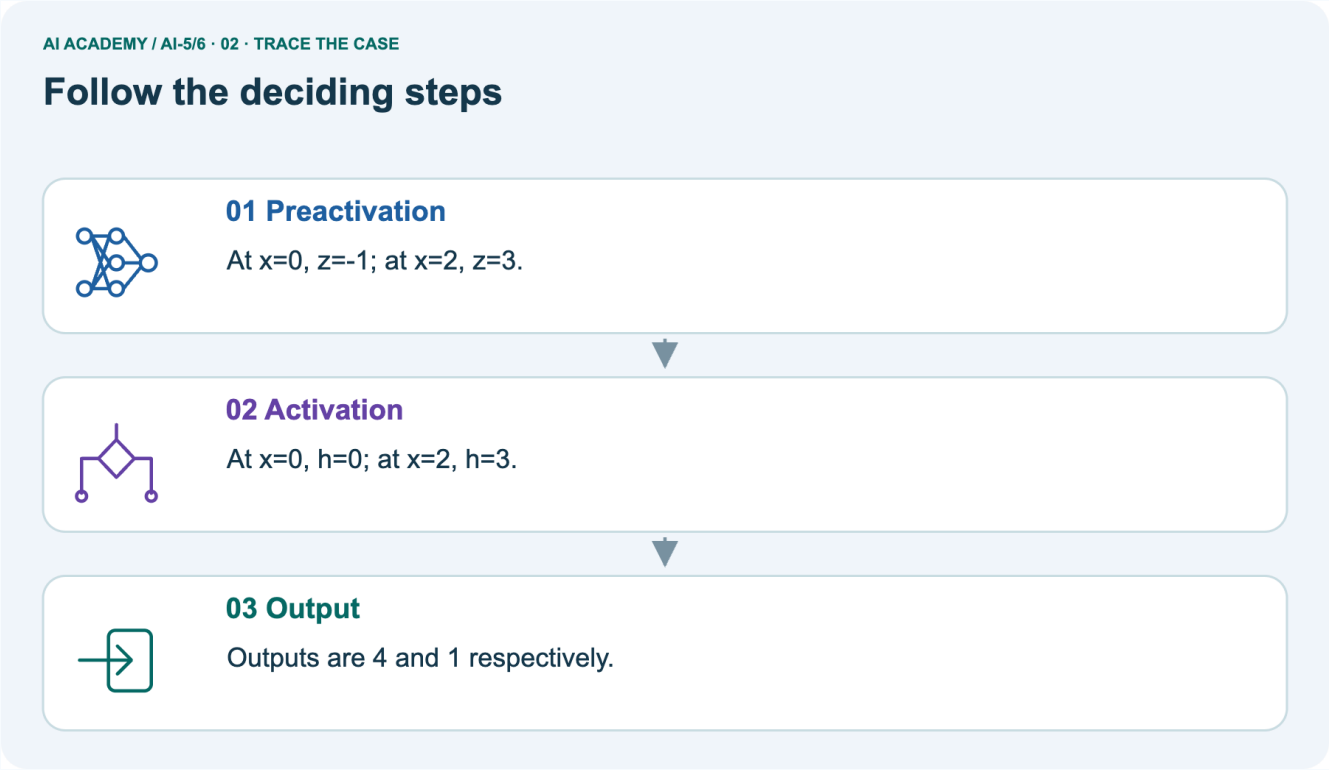
Follow the labelled picture with the matching explanation. Open the vector diagram

Audit a forward pass by writing every value at its correct stage. Supplied facts: Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Preactivation: At x=0, z=-1; at x=2, z=3. → Activation: At x=0, h=0; at x=2, h=3. → **Output**: **Output**s are 4 and 1 respectively.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Correct a trace from the earliest error

AI ACADEMY / AI-5/6 · 03 · INSPECT THE LESSON MODEL

Correct a trace from the earliest error

Trace stage at x is 0	Expected value	Submitted value
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

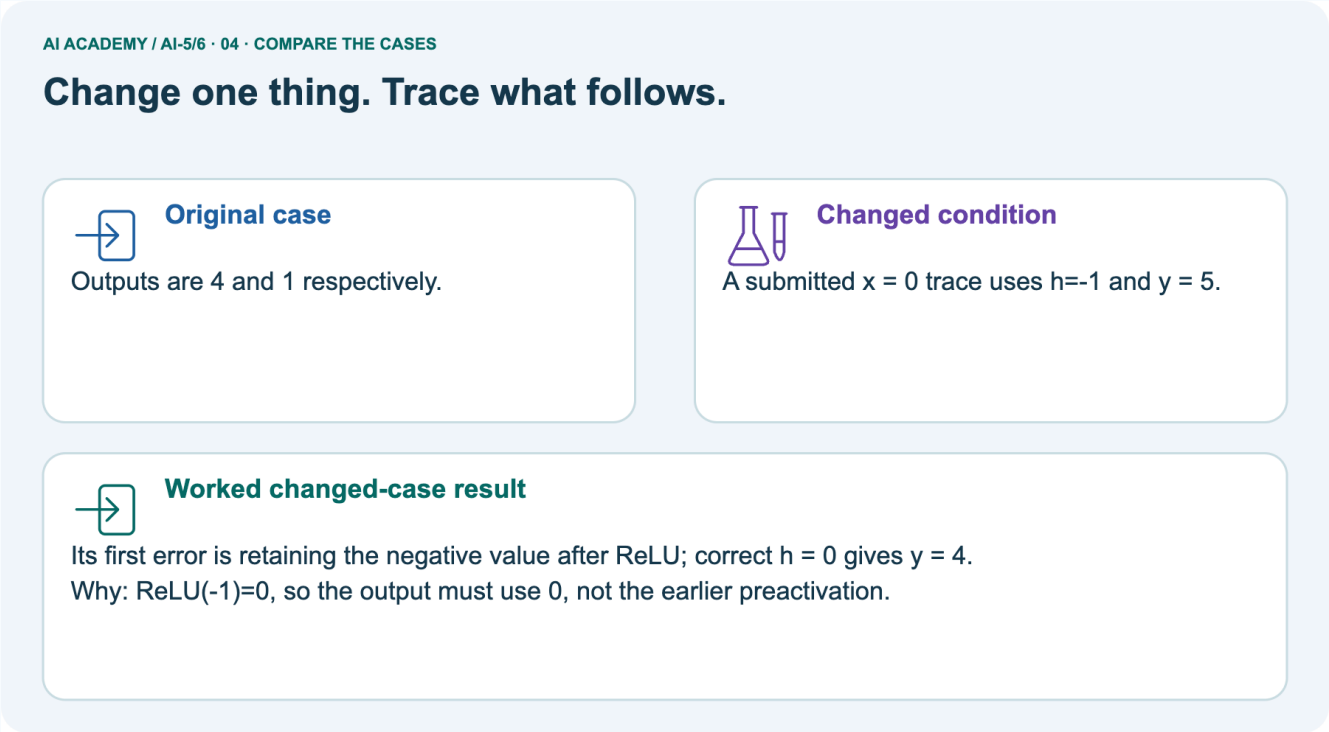
The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Follow the labelled picture with the matching explanation. Open the vector diagram

Correct a **trace** from the earliest error. Scores | [-1, 2] | [-1, 2]. ReLU activations | [0, 2] | [-1, 2]. **Output** | -3 | -4

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



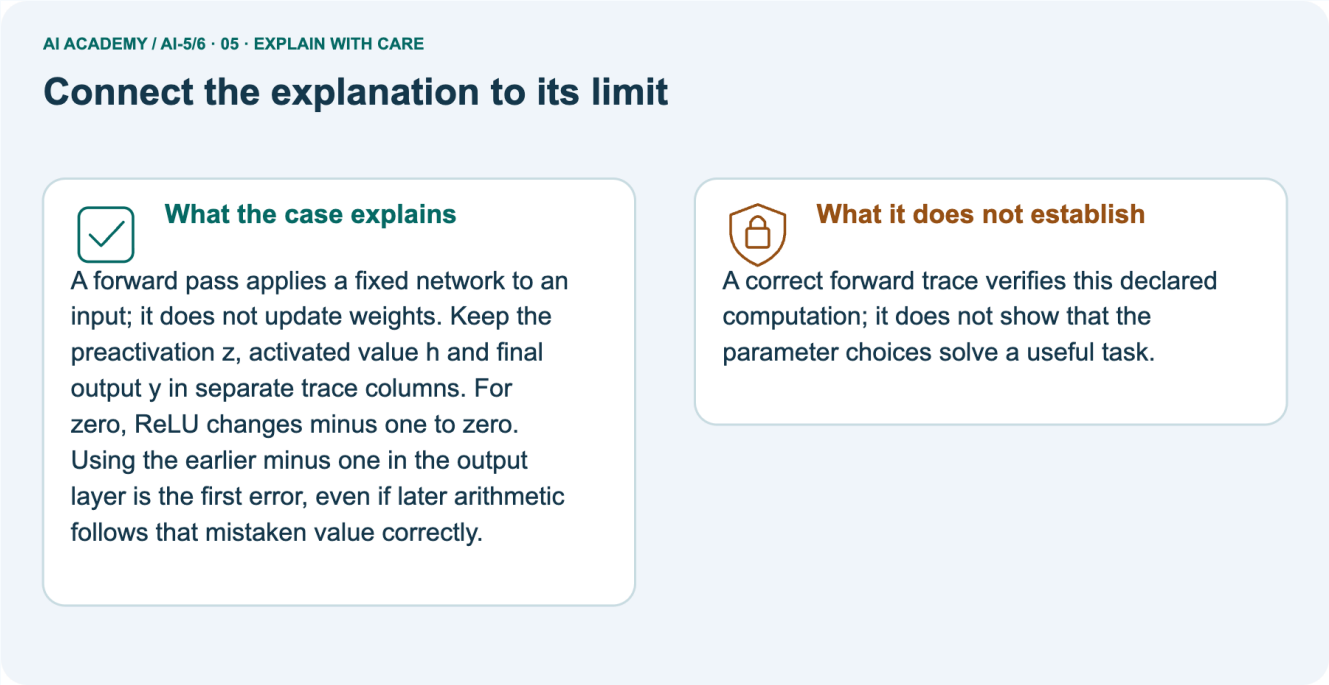
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Output**s are 4 and 1 respectively. Changed condition: A submitted x = 0 **trace** uses h=-1 and y = 5. Expected result: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

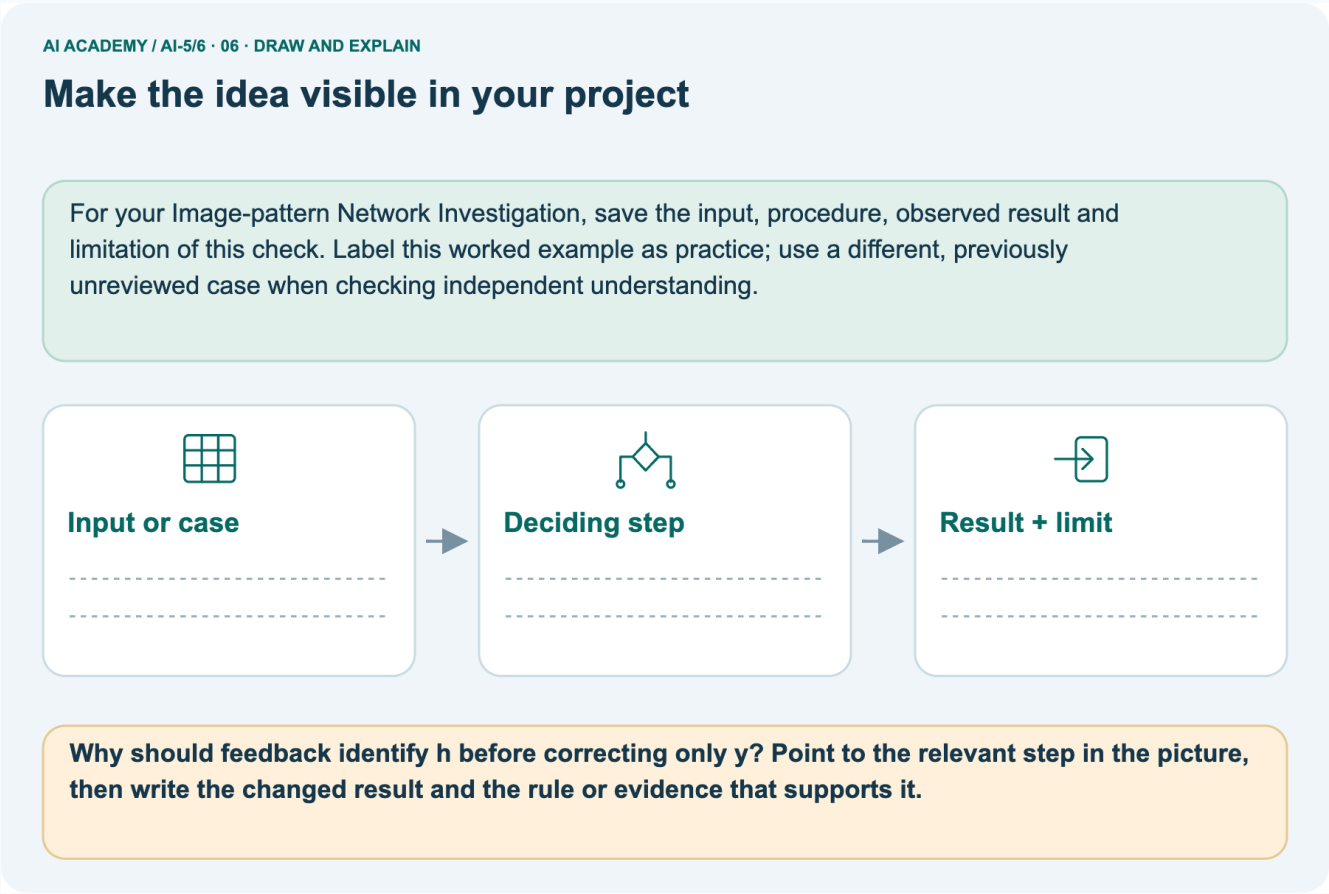
A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Limit: A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project **trace** and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.